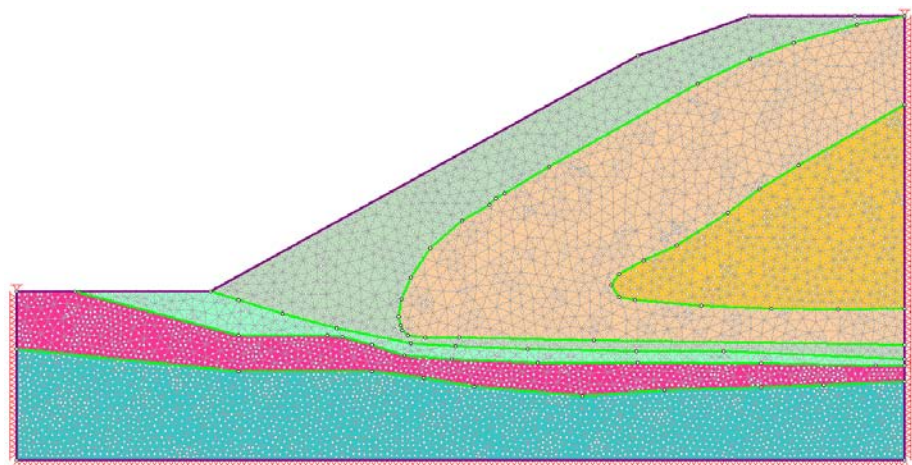


Probabilistic Slope Stability Analysis

This tutorial will demonstrate how to carry out a probabilistic finite element slope stability analysis with *Phase2* using the SSR (shear strength reduction) method in combination with the point estimate method of probabilistic analysis. Interpretation of results will be discussed and a comparison with *Slide* limit equilibrium slope stability results will be carried out.

Topics covered:

- Point estimate method of probabilistic analysis
- SSR slope stability analysis
- Probability of failure
- Component files
- Comparison with *Slide* results



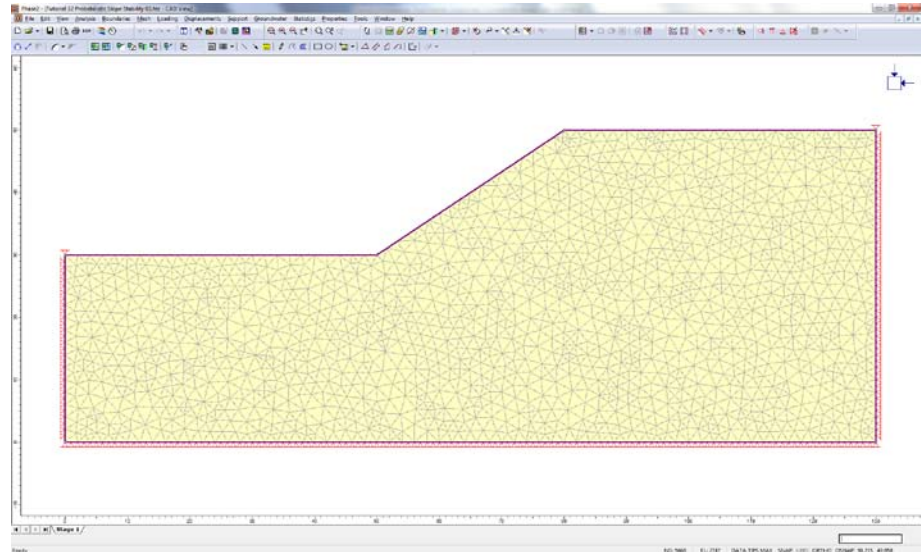
We will run two different models in this tutorial:

- a simple homogeneous slope with two random variables
- a complex multi-material slope with six random variables

Simple Homogeneous Slope

First we will run a probabilistic slope stability analysis on a simple homogeneous slope model.

From the *Phase2* main menu select File > Recent Folders > Tutorials Folder. Open the file **Tutorial 32 Probabilistic Slope Stability 01.fe2**.



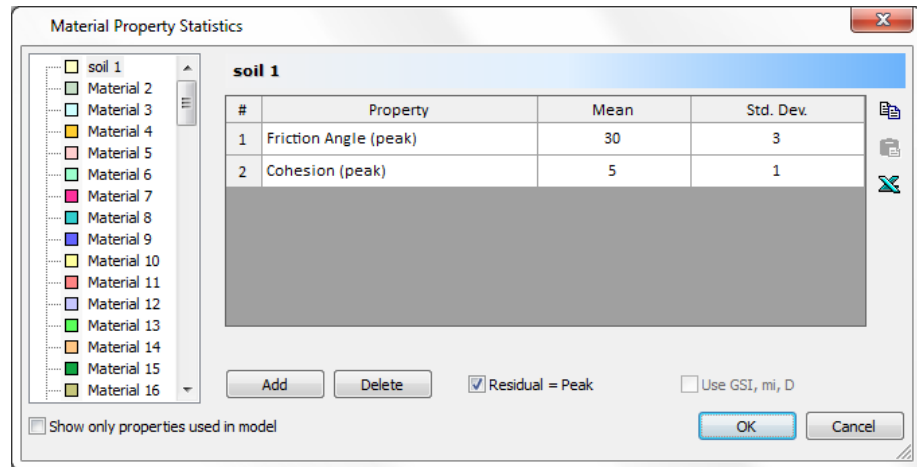
Project Settings

Let's have a look at the Project Settings.

1. Select **Project Settings** from the **Analysis** menu.
2. Select the Statistics page in the Project Settings dialog. Notice that the Probabilistic Analysis checkbox is selected.
3. Select the Strength Reduction page. Notice that the Determine Strength Reduction Factor checkbox is selected.
4. When both of these checkboxes are selected, this allows you to carry out a probabilistic slope stability analysis.
5. Since we are not changing any Project Settings select Cancel.

Material Property Statistics

For this model two random variables have been defined – the cohesion and friction angle of the slope material. Random variables are defined using the options in the Statistics menu. Select **Material Properties** from the **Statistics** menu.



As you can see in the Material Property Statistics dialog the defined random variables are:

- Friction Angle (Mean = 30, Standard Deviation = 3)
- Cohesion (Mean = 5, Standard Deviation = 1)

Since the **Residual = Peak** checkbox is selected this means that the material is perfectly plastic (i.e. residual strength = peak strength) so it is not necessary to define residual strength parameters.

Select **Cancel** in the dialog since the variables are already defined.

Compute

Since the model is already fully defined we can go ahead and Compute.

Select **Compute** from the toolbar or the Analysis menu.

When you compute a probabilistic analysis model in *Phase2* the following will occur:

- First, the model is computed using the MEAN values of all variables. This is the same analysis that would be carried out if you were NOT using the probabilistic analysis option (i.e. a regular deterministic analysis).
- Then the probabilistic analysis is computed. For the point estimate method, this consists of running the analysis for all possible combinations of random variable point estimates. Since we have two random variables defined for this model, 4 separate analyses will be computed ($2^2 = 4$) i.e. the analysis will be run 4 times using the following combinations of cohesion and friction angle.

Cohesion	Friction Angle
4	27
4	33
6	27
6	33

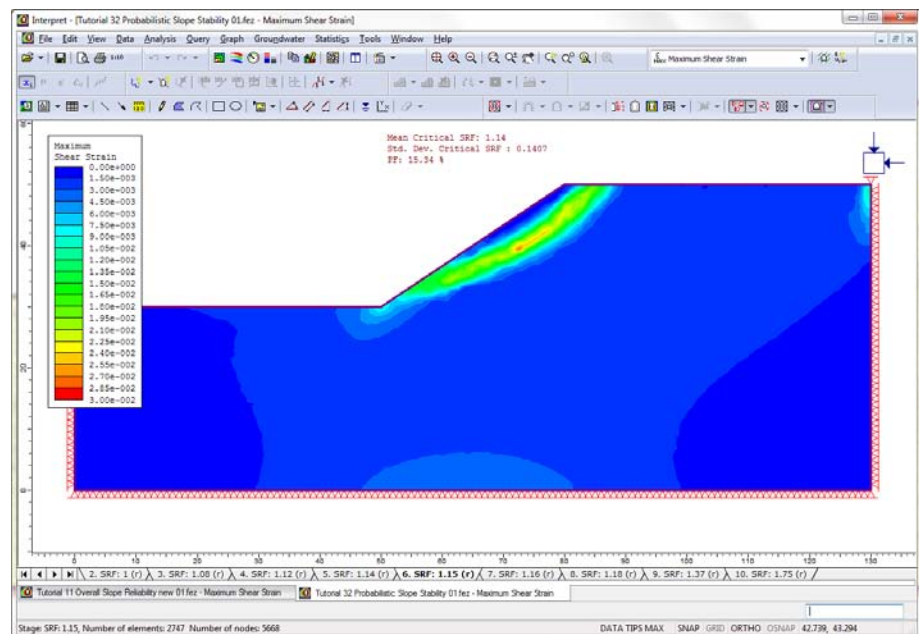
Remember that the random variable point estimates are given by plus or minus one standard deviation from the mean value.

Since we are running an SSR slope stability analysis in conjunction with a probabilistic analysis, remember that each analysis run generated by the probabilistic analysis requires a complete SSR slope stability analysis, using a new set of random variable inputs.

If you have a fast computer the analysis should take a few minutes to run. When the Compute is finished we will examine the results of the probabilistic SSR analysis.

Interpret

Select **Interpret** from the Analysis menu. Note: to see the figure below select the tab SRF = 1.15 to highlight the zone of maximum shear strain.



The primary results of the probabilistic SSR slope stability analysis are listed at the top center of the view:

- Mean Critical SRF
- Std. Dev. Critical SRF
- PF (probability of failure)

The Mean Critical SRF is the mean critical strength reduction factor (i.e. safety factor) obtained from the probabilistic analysis runs.

Select the **Info Viewer**, scroll down to the Strength Reduction Factor Statistics section and you can see how this number is generated.

Strength Reduction Factor Statistics

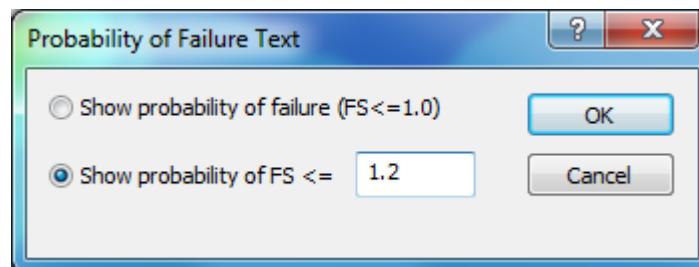
Probability of Failure: 15.34 %
 Critical SRF mean value: 1.14
 Critical SRF std. dev. value: 0.1407

File #	Critical SRF
1	1.3
2	1.06
3	1.22
4	0.995

The Mean Critical SRF is simply the average of the values obtained from the four SSR analysis runs generated by the probabilistic analysis (i.e. $(1.3 + 1.06 + 1.22 + .995) / 4 = 1.14$). The standard deviation of the critical SRF is the standard deviation of these four values. The probability of failure is computed by assuming a normal distribution for all input and output random variables, and calculating the probability of the critical SRF being less than 1.

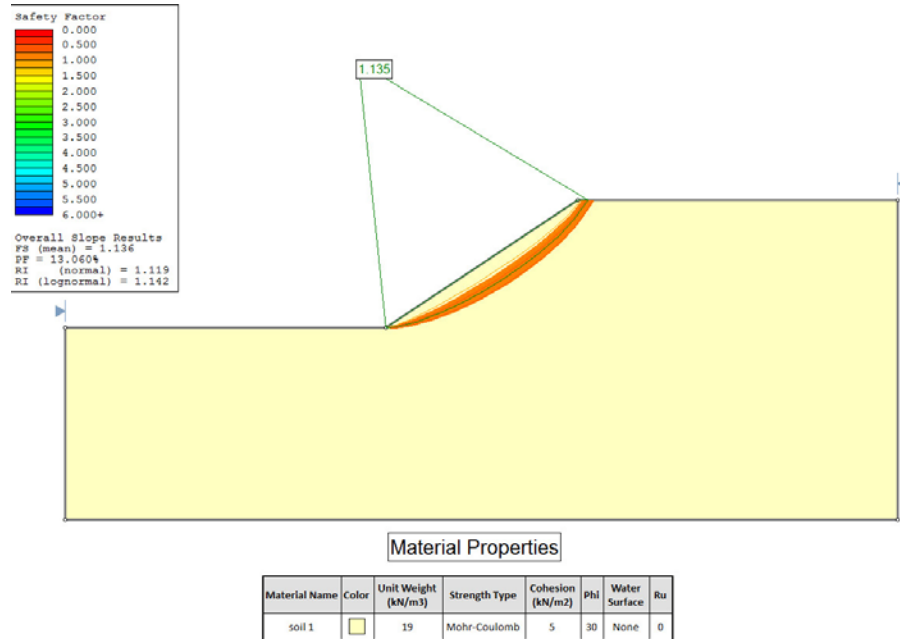
Close the Info Viewer view.

By default a value of critical SRF = 1 is used to calculate the probability of failure. If you wish, you can define a value other than SRF = 1 as the definition of “failure”. To do this, select Statistics > Probability of Failure and enter a value. If you change this value you will obtain a different probability of failure as displayed in the text at the top of the view. This is left as an optional exercise to experiment with.



Comparison with *Slide* results

For purposes of comparison, this model was exported to the limit equilibrium slope stability program *Slide*. A probabilistic analysis was run using the Overall Slope method of probabilistic analysis. The results are summarized below.



Program Name	Probability of Failure	Mean Factor of Safety	Standard Deviation F.S.
Phase2 8.0	15.3	1.14	0.14
Slide 6.0	13.1	1.14	0.12

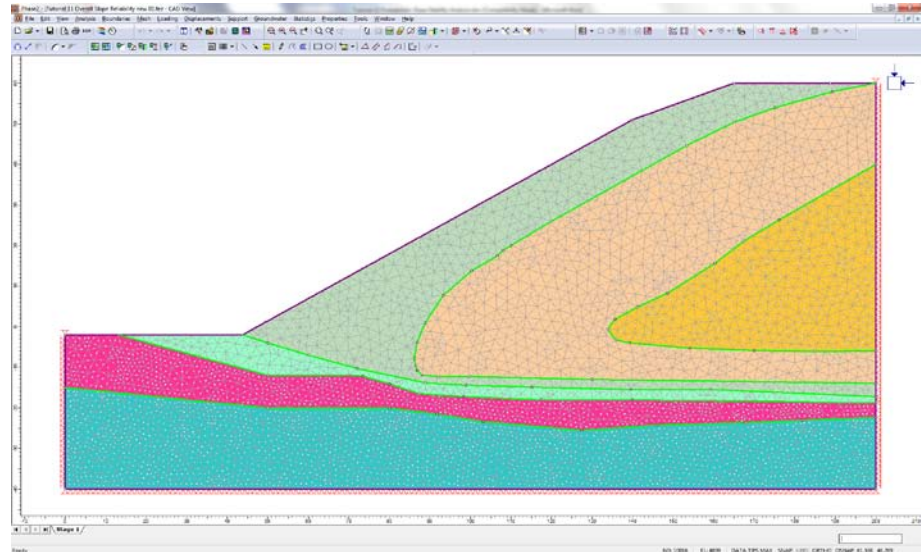
These results show that both *Phase2* and *Slide* give nearly identical results for a probabilistic slope stability analysis of this simple homogenous slope model.

If you have the *Slide* program, you can run the file **Tutorial 32 Probabilistic Slope Stability 01.slim** to verify these results. You will find this file in the Phase2 Examples > Tutorials folder.

Complex Multi-Material Slope

Now let's run a probabilistic slope stability analysis on a more complex slope model with six random variables.

From the *Phase2* main menu select File > Recent Folders > Tutorials Folder. Open the file **Tutorial 32 Probabilistic Slope Stability 02.fez**.



Generate Component Files

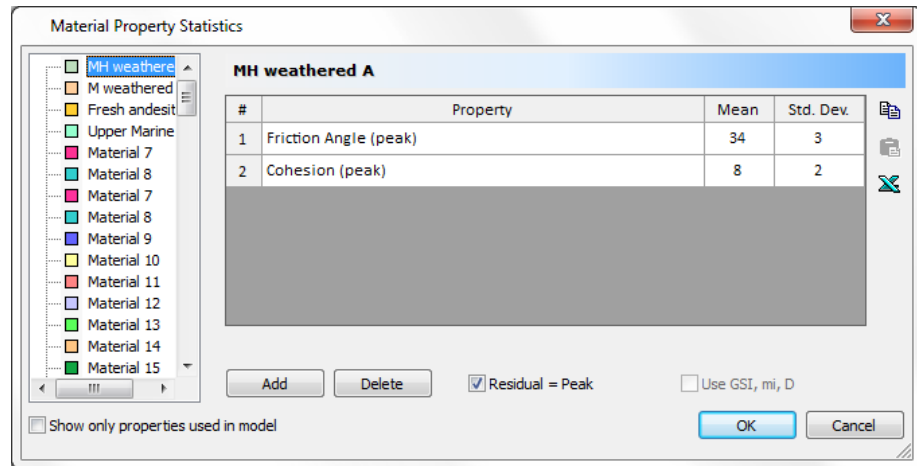
Before we do anything with this model, we will generate the probabilistic component files by selecting **File > Save**. This may take some time, and you will see a progress bar at the bottom of the screen.

The reason for doing this, is because this model with 6 random variables, generates 64 component files. For this rather complex model, the 64 component files require over 20 MB of space, even without analysis results. In order to save space, these component files were not included with the *Phase2* installation, for this particular file.

NOTE: if you examine the *Phase2* Tutorials folder where the tutorial files are stored, you will notice sub-folders which have the same names as the probabilistic tutorial files. These sub-folders are used to store the probabilistic component files for each of the *Phase2* probabilistic example files. The number of files in each sub-folder corresponds to the number of random variables in the master file (e.g. 2 variables = 4 files, 3 variables = 8 files, 4 variables = 16 files etc).

Material Property Statistics

Now take a look at the material property random variables which have been defined. Select **Material Properties** from the **Statistics** menu.



If you select the first four materials from the list at the left of the dialog, you will see that six random variables have been defined (i.e. friction angle and cohesion for three of the four materials).

Six random variables will require $2^6 = 64$ separate analyses using the point estimate method of probabilistic analysis.

Select Cancel in the dialog.

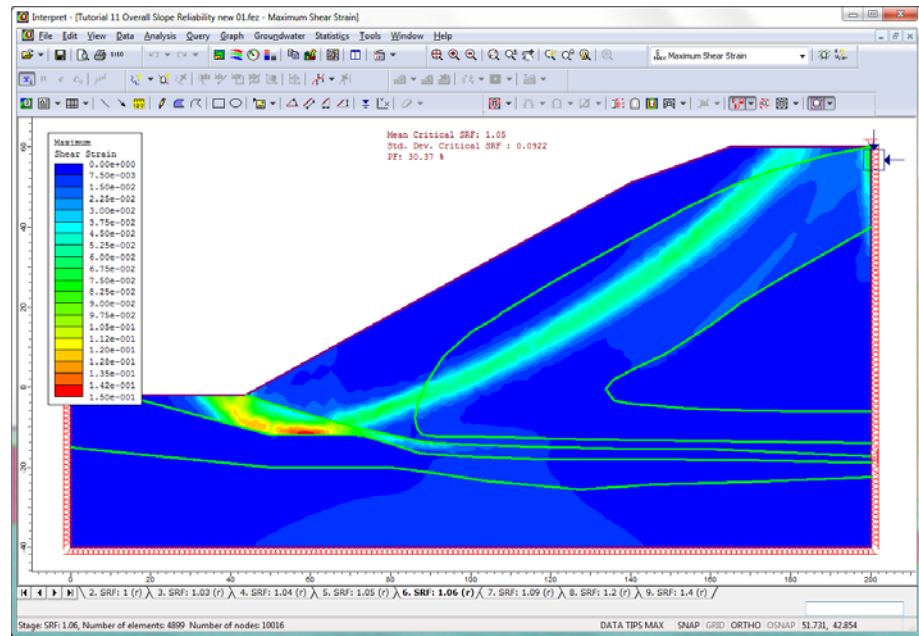
Compute

Because this probabilistic analysis will require 64 separate analysis runs, and each run is an SSR finite element slope stability analysis of a relatively complex model, this analysis will take a significant amount of computation time.

If you have a fast computer, it may take about 3 hours. If you have a slow machine, you may need to run this analysis overnight.

Interpret

Once the probabilistic analysis has been computed, you should see the following results. Note: to see the figure below select the tab $SRF = 1.06$ to highlight the zone of maximum shear strain.

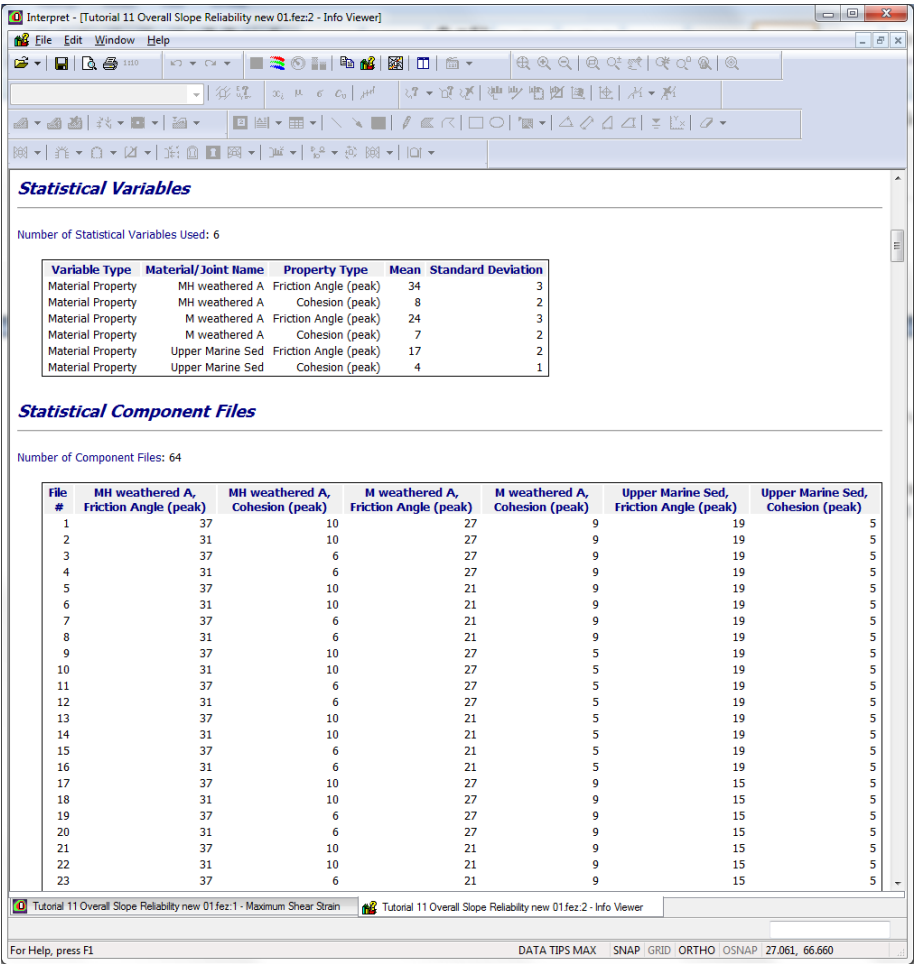


The primary results of the probabilistic SSR slope stability analysis are listed at the top center of the view:

- Mean Critical SRF
- Std. Dev. Critical SRF
- PF (probability of failure)

The definitions of these are discussed in the previous example.

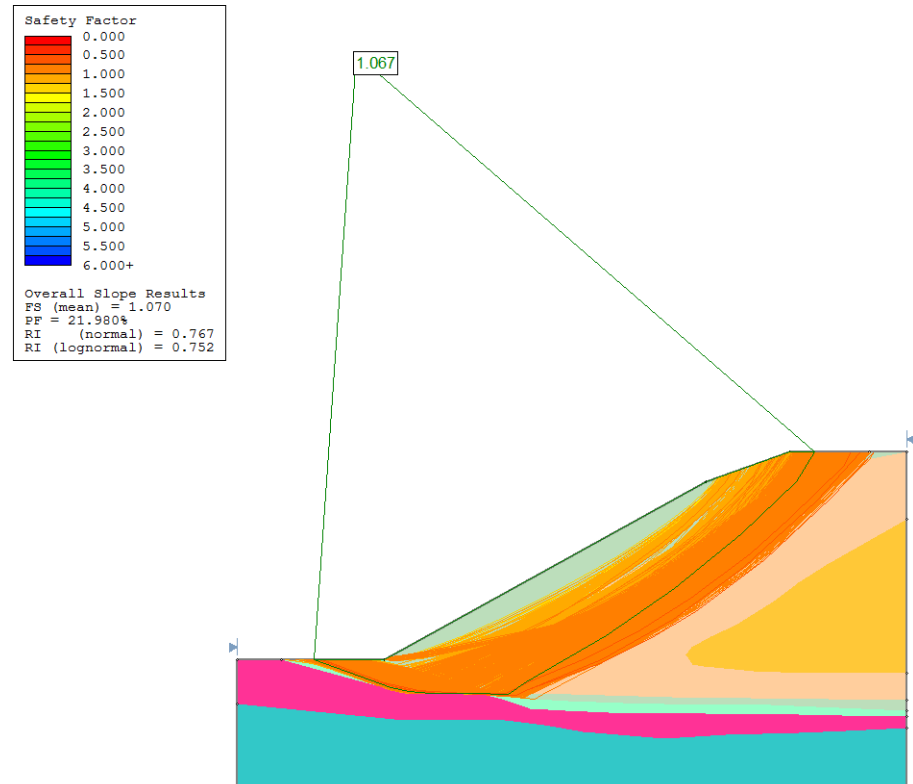
Select the **Info Viewer** and examine the summary of statistical input and output data for this model. You will see the input and output for the 64 component files of the probabilistic analysis.



Close the Info Viewer.

Comparison with *Slide* results

For purposes of comparison, this model was exported to the limit equilibrium slope stability program *Slide*. A probabilistic analysis was run using the Overall Slope method of probabilistic analysis. The results are summarized below.



Program Name	Probability of Failure	Mean Factor of Safety	Standard Deviation F.S.
Phase2 8.0	30.4	1.05	0.092
Slide 6.0	22.0	1.07	0.091

In this case the Mean Factor of Safety and Standard Deviation computed by *Phase2* and *Slide* are nearly identical. The Probability of Failure computed by *Phase2* (30.4) is significantly higher than *Slide* (22.0). However this is primarily due to the fact that the mean safety factor is very close to 1, therefore small differences in the mean value can result in a substantially different probability of failure, since the definition of failure is safety factor = 1.

If you have the *Slide* program, you can run the file **Tutorial 32 Probabilistic Slope Stability 02.slim** to verify these results. You will find this file in the Phase2 Examples > Tutorials folder.

Summary

Probabilistic slope stability analysis can be easily carried out using the point estimate method of probabilistic analysis and the shear strength reduction (SSR) slope stability analysis available in the finite element program *Phase2*.

The results computed by *Phase2* (mean safety factor and probability of failure) have been compared to limit equilibrium slope stability analysis results computed by *Slide* and found to be in good agreement.